Fall - 2023

Sample Ques for Mid

1] (a) $2x - 5y = 7$

$\Rightarrow 5y = 2x - 7$

$\Rightarrow y = \frac{2x-7}{5}$

$\therefore f(x) = \frac{2x-7}{5}$

There is no value of x so that we'll get two or more values of $f(x)$. Thus it is a function.

(b)

$(y+3)^3+1 = 2x$

$\Rightarrow (y+3)^3 = 2x-1$

$\Rightarrow y+3 = \sqrt[3]{2x-1}$

$\Rightarrow y = \sqrt[3]{2x-1} - 3$

$\therefore f(x) = \sqrt[3]{2x-1} - 3$

There is no ... . Thus $f(x)$ is a function

(c)

$x^2+(y-3)^2 = 5$

$\Rightarrow (y-3)^2 = 5-x^2$

$\Rightarrow y-3 = \pm\sqrt{5-x^2}$

$\Rightarrow y = 3 \pm \sqrt{5-x^2}$

$\therefore f(x) = 3 \pm \sqrt{5-x^2}$

Here,

$f(1) = 3 \pm \sqrt{5-1^2}$

$= 3 \pm 2$

$\Rightarrow f(1) = 5, 1$; that is two values of 'y'

$\therefore f(x)$ is not a function

(d) $2x - |y| = 0$

$\Rightarrow |y| = 2x$

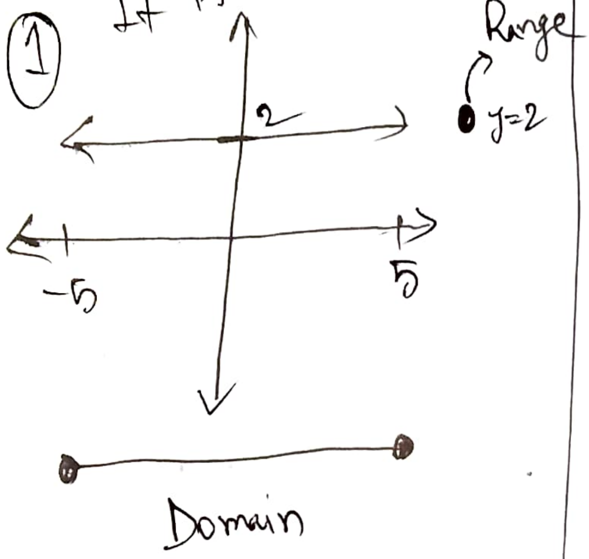
$\Rightarrow y^2 = (2x)^2$

$\Rightarrow y = \pm 2x \Rightarrow f(x) = \pm 2x$

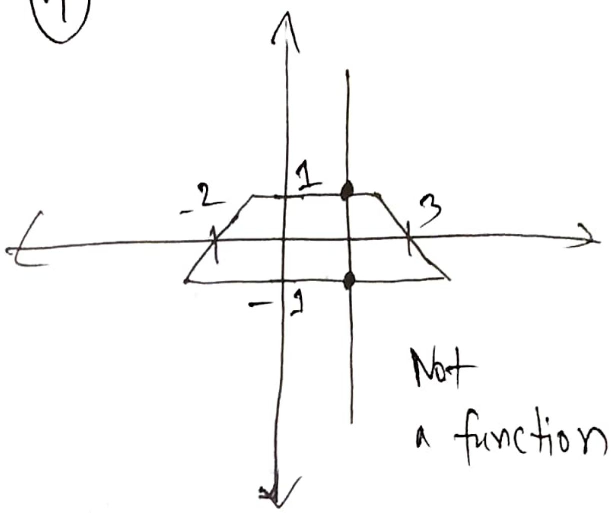
$\therefore f(1) = 2, -2 \Rightarrow f(x)$ is not a function

2)

(1) It is a function

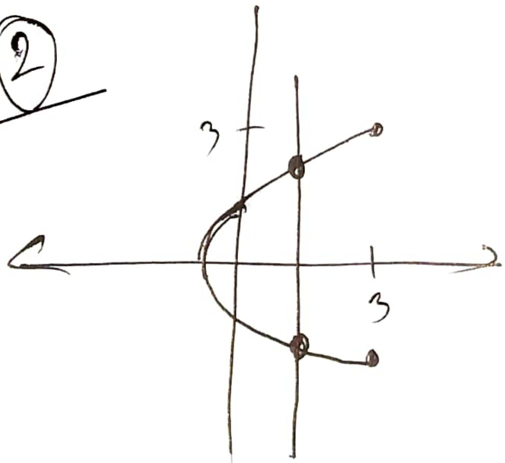


(4)



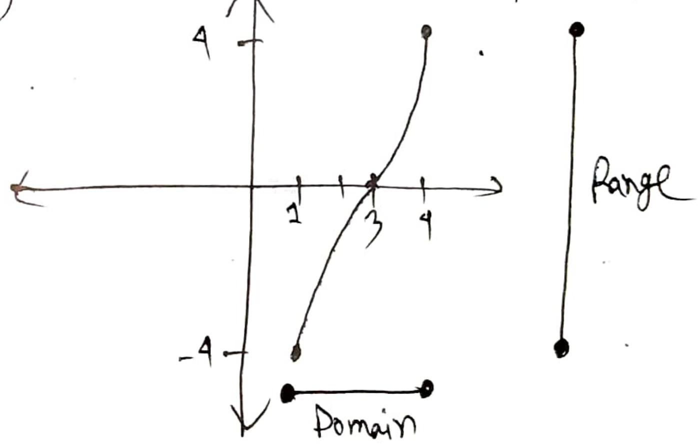
Not a function

(2)

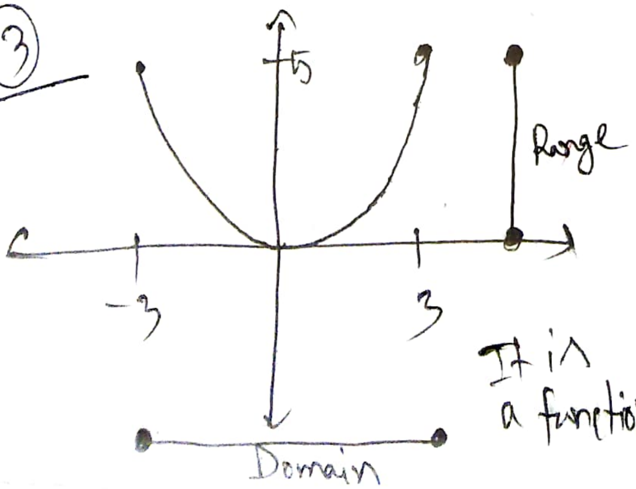


Not a function

(5) It is a function

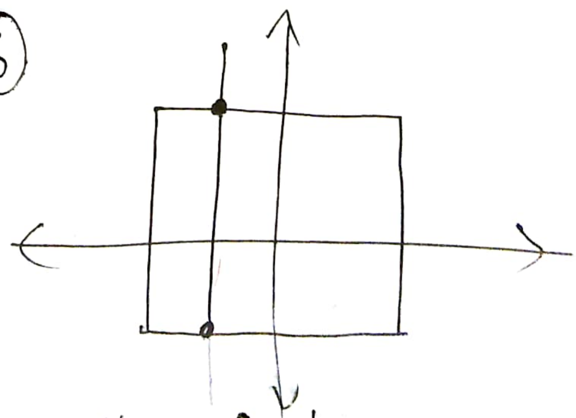


(3)



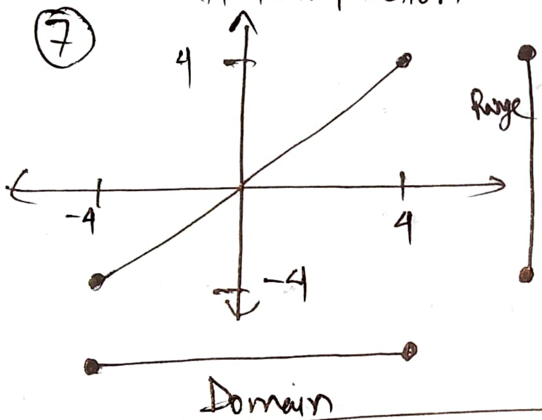
It is a function

(6)



Not a function

7 It is a function



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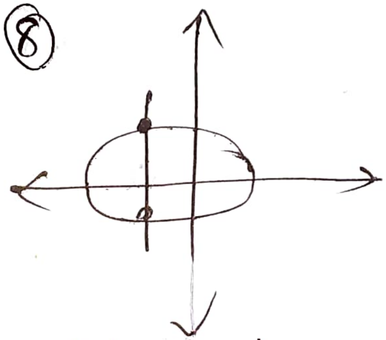
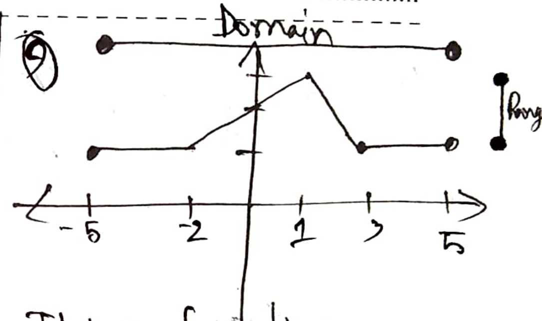


Not a function

9



It is a function

3)

(a) $f(-4) = -2, \quad g(3) = 4$

(b) $f(-3) < g(-3)$

(c) $f(x) = g(x) \Rightarrow x = -2, 2$

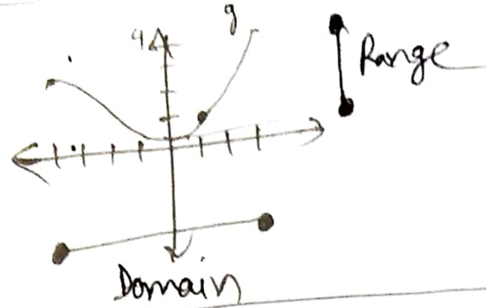
(d) $f(x) \leq g(x) \Rightarrow x \in [-4, -2) \cup (2, 4]$

(e) $f(x) = -1 \Rightarrow \cancel{x = 4}\ x = -3, 4$

(f) g is decreasing in $[-4, 0)$

(g) Dom $f = [-4, 4]$, Range $f = [-2, 3]$

(h)



(i) f is increasing on $[-4, 0)$

4) (a) $f(x) = \dfrac{x+4}{x^2-9}$. Here $x^2-9 \neq 0 \Rightarrow x \neq \pm 3$

$\therefore$ Dom $f = \mathbb{R} - \{-3, 3\} = (-\infty, -3) \cup (-3, 3) \cup (3, \infty)$

(b) $g(x) = \dfrac{x^2+1}{x^2+4x-21}$. Here $x^2+4x-21 \neq 0 \Rightarrow x \neq -7, 3$

$\therefore$ Dom $g = \mathbb{R} - \{-7, 3\} = (-\infty, -7) \cup (-7, 3) \cup (3, \infty)$

(c) $h(x) = \sqrt{x^2-4x-5}$. Here,

	$x<-1$	$-1<x<5$	$x>5$
$(x-5)$	$(-)$	$(-)$	$(+)$
$(x+1)$	$(-)$	$(+)$	$(+)$
	$(+)$	$(-)$	$(+)$

$x^2-4x-5 \geq 0 \Rightarrow (x-5)(x+1) \geq 0$

$\Rightarrow x \leq -1$ ~~and~~ or $x \geq 5$

$\therefore$ Dom $h = (-\infty, -1] \cup [5, \infty)$

(d) $k(x) = \sqrt{3-x} - \sqrt{2+x}$

Here, $3-x \geq 0$ and $2+x \geq 0$

$\Rightarrow x \leq 3$ $\Rightarrow x \geq -2$

i.e. $-2 \leq x \leq 3$

$\therefore$ Dom $k = [-2, 3]$

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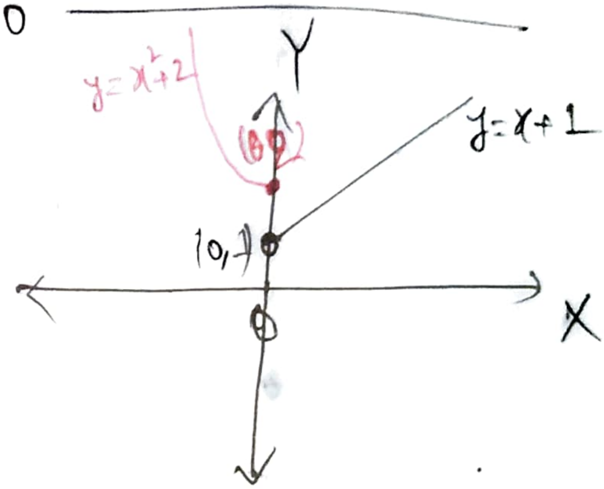
5 (a)

$f(-4) = (-4)^2 + 2 = 18$

$f(0) = 0^2 + 2 = 2$

$f(2) = 2+1 = 2$

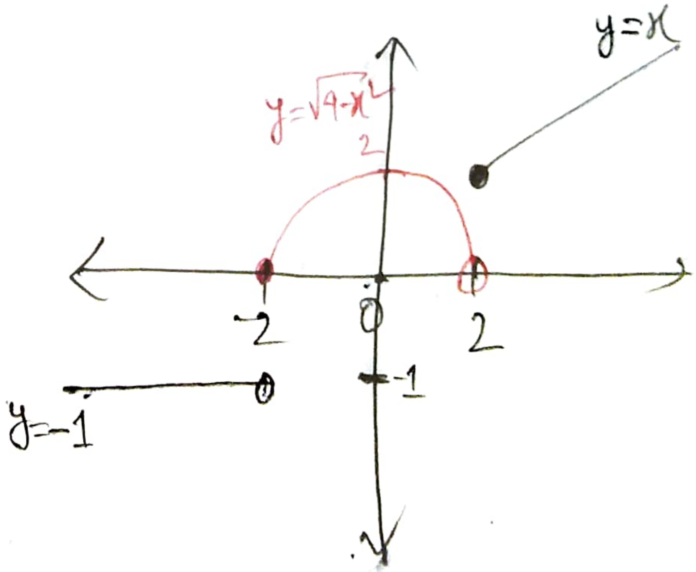


(b)

$f(-4) = -1$

$f(0) = \sqrt{4-0^2} = 2$

$f(2) = 2$

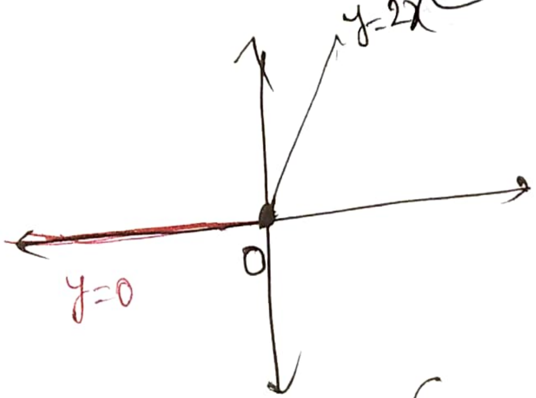


Hint $y = \sqrt{4-x^2} \Rightarrow y^2 = 4-x^2$

$\therefore x^2 + y^2 = 4 = 2^2$

i.e. upper semi-circle with radius 2

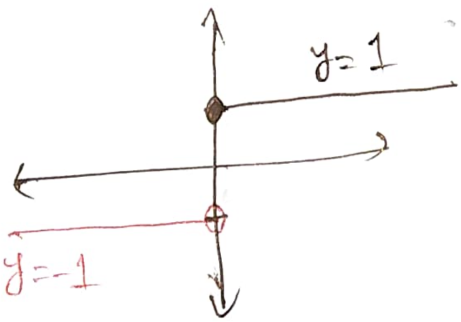
6) (a) $f(x) = x+|x| = \begin{cases} x+x \text{ or, } 2x & ; x \geq 0 \\ x-x \text{ or, } 0 & ; x<0 \end{cases}$



$\text{Dom} f = \mathbb{R} = (-\infty, \infty)$

$\text{Range} f = [0, \infty)$

(b) $f(x) = \dfrac{|x|}{x} = \begin{cases} \frac{x}{x} \text{ or, } 1 & ; x \geq 0 \\ \frac{-x}{x} \text{ or, } -1 & ; x<0 \end{cases}$



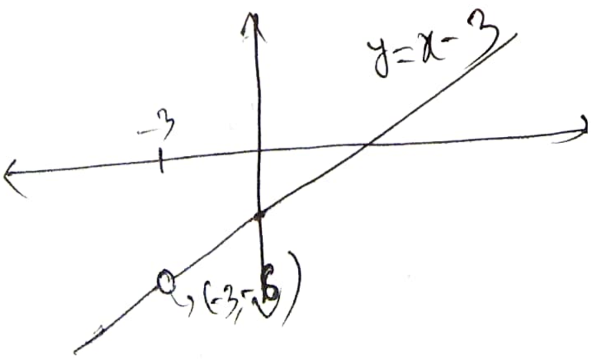
$\text{Dom} f = \mathbb{R}$

$\text{Range} f = \{-1, 1\}$

(c) $f(x) = \dfrac{x^2-9}{x+3} = \dfrac{(x+3)(x-3)}{(x+3)} = x-3 \quad ; x \neq -3$

$\Rightarrow y \neq -3-3$

$\therefore y \neq -6$



$\text{Dom} f = \mathbb{R} - \{-3\}$

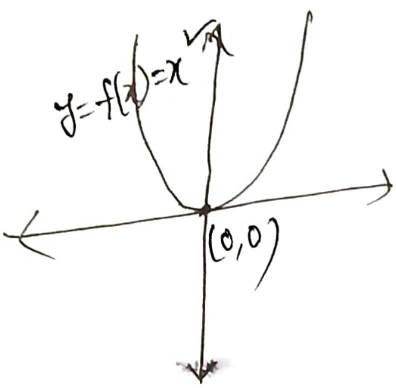
$= (-\infty, -3) \cup (-3, \infty)$

$\text{Range} f = \mathbb{R} - \{-6\}$

$= (-\infty, -6) \cup (-6, \infty)$

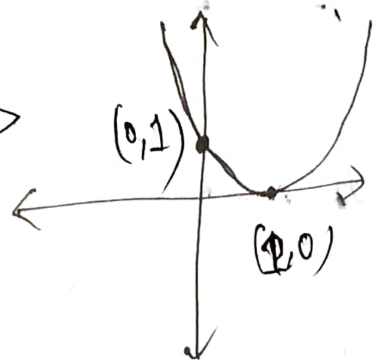
(d) $y = 2(x-1)^2 + 3$

$y = f(x) = x^2$



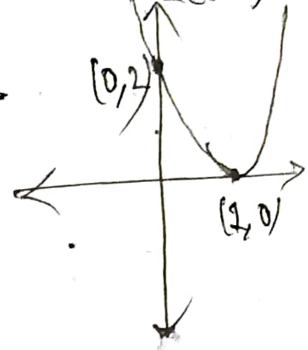
$\Rightarrow$

$y = f(x-1) = (x-1)^2$



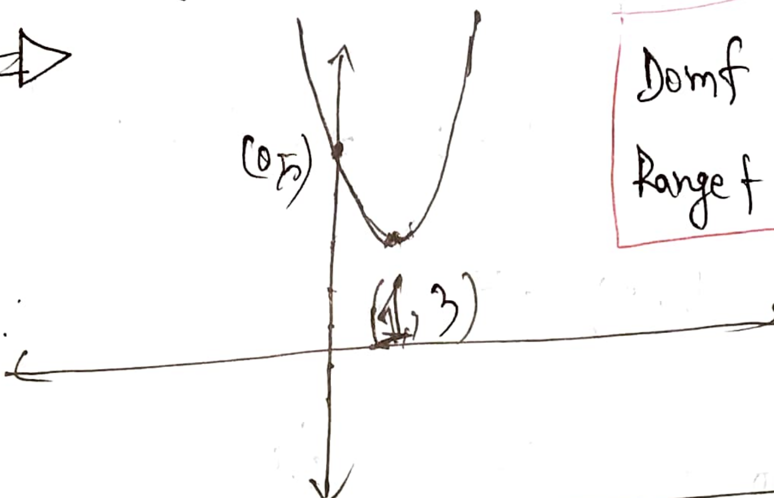
$\Rightarrow$

$y = 2f(x-1)$
$= 2(x-1)^2$



$y = 2f(x-1) + 3 = 2(x-1)^2 + 3$

$\Rightarrow$

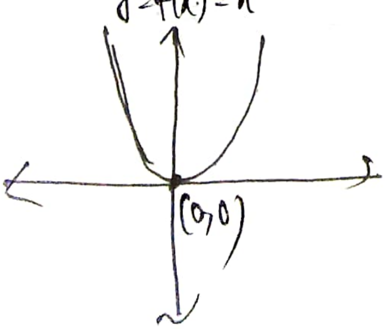


$$\boxed{\begin{array}{l} \text{Dom} f = \mathbb{R} = (-\infty, \infty) \\ \text{Range } f = [3, \infty) \end{array}}$$

(e) $y = 4x - x^2 = -(x^2 - 4x) = -(x^2 - 2 \cdot x \cdot 2 + 2^2 - 2^2)$

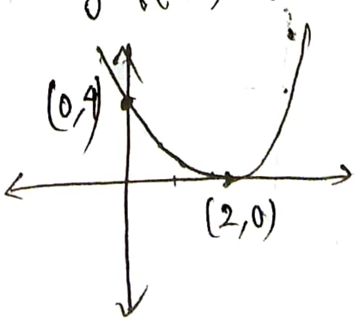
$\Rightarrow \boxed{y = -(x-2)^2 + 4}$

$y = f(x) = x^2$



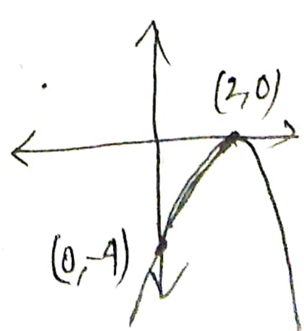
$\Rightarrow$

$y = f(x-2) = (x-2)^2$



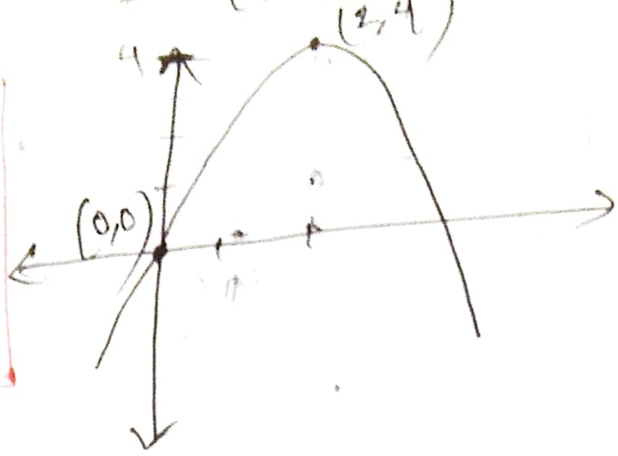
$\Rightarrow$

$y = -f(x-2)$
$= -(x-2)^2$



$y = -f(x-2)+4$

$\Longrightarrow \quad = -(x-2)^2+4$



Dom f = $\mathbb{R}$

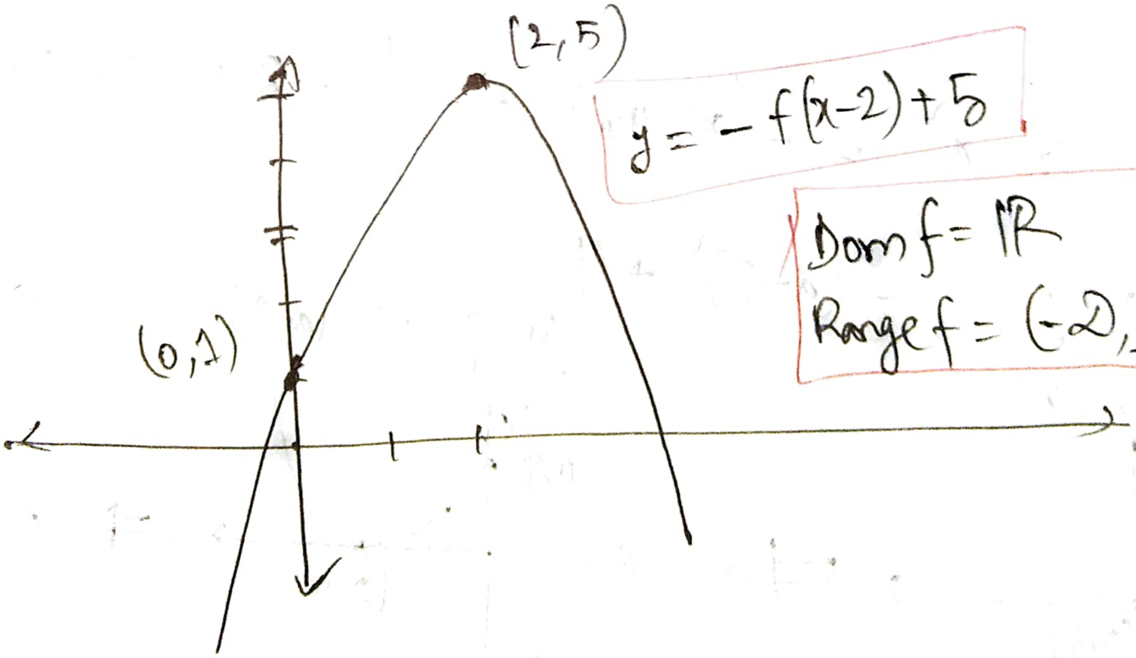
~~Dom~~

Range f = $(-\infty, 4]$

(f) $y = 1+4x-x^2 = -(x^2-4x-1) = -(x^2-2\cdot x\cdot 2+2^2-2^2-1)$

$\Rightarrow y = -(x-2)^2+5$

It is similar to (e). The final figure is,



$y = -f(x-2)+5$

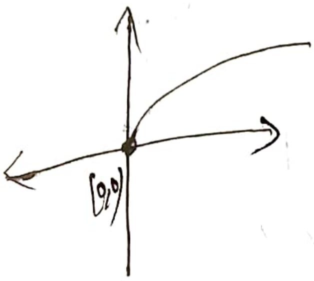
Dom f = $\mathbb{R}$

Range f = $(-\infty, 5]$

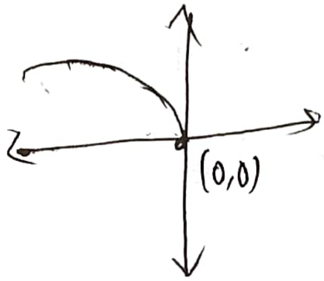
(g) $y = 2 - \sqrt{3-x}$

$-x+3=0 \Rightarrow x=3$

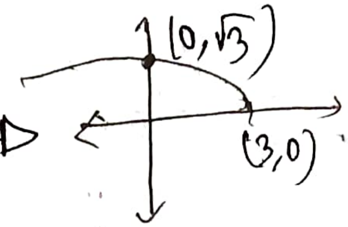
$y=f(x)=\sqrt{x}$



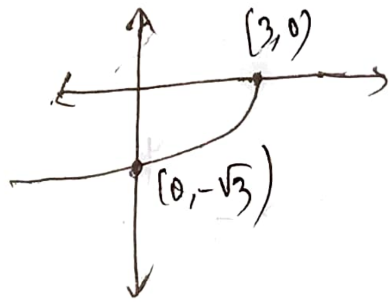
$\Rightarrow$ $y=f(-x)=\sqrt{-x}$



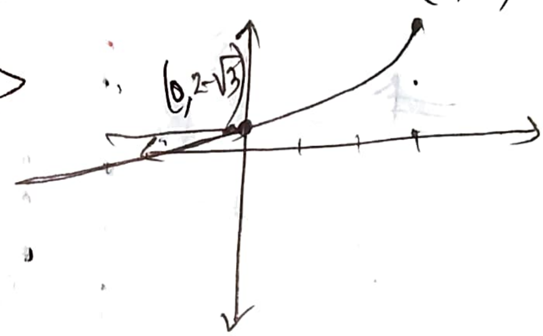
$\Rightarrow$ $y=f(-x+3)=\sqrt{-x+3}$



$\Rightarrow$ $y=-f(-x+3)=-\sqrt{3-x}$



$\Rightarrow$ $y=-f(-x+3)+2 = 2-\sqrt{3-x}$



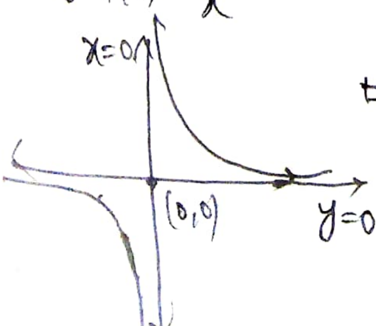
$$\left.\begin{aligned} \text{Dom} f &= (-\infty, 3] \\ \text{Range} f &= (-\infty, 2] \end{aligned}\right\} \text{using fig.}$$

(h) $y = \frac{x}{x+1} = \frac{x+1-1}{x+1} = 1 - \frac{1}{x+1}$

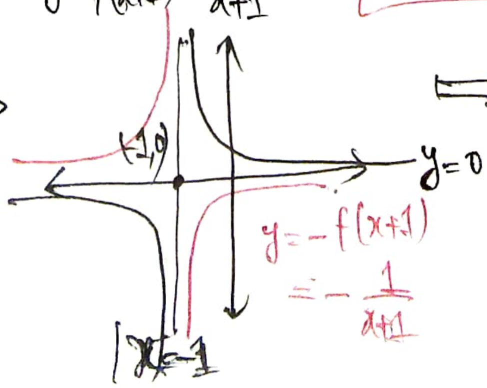
Dom $f = \mathbb{R} - \{-1\}$
Range $f = \mathbb{R} - \{1\}$

$y=1-f(x+1)$

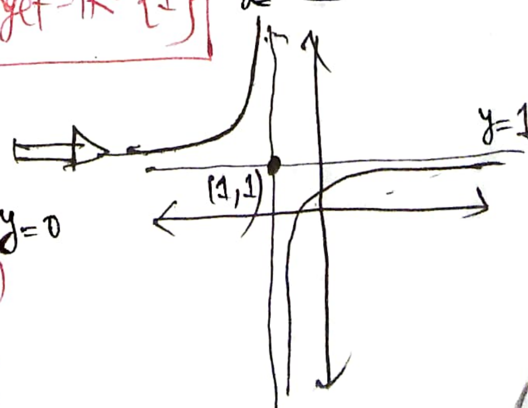
$y=f(x)=\frac{1}{x}$



$\Rightarrow$ $y=f(x+1)=\frac{1}{x+1}$

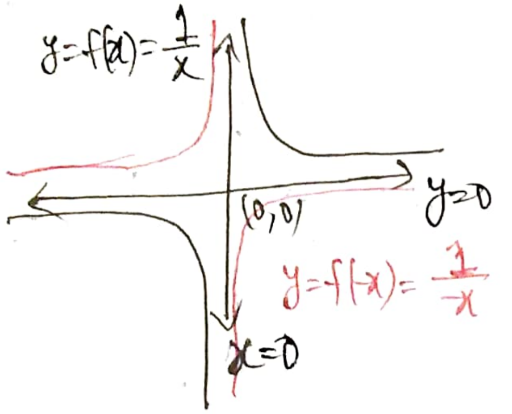


$\Rightarrow$



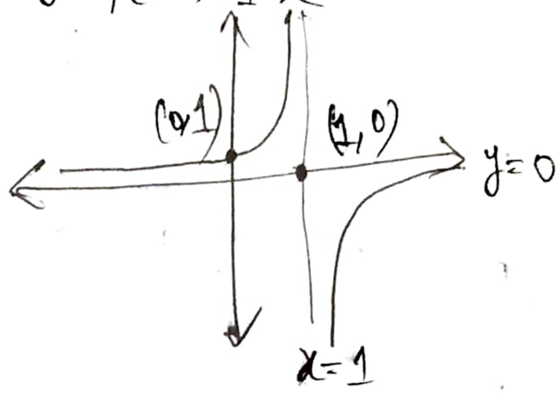
(i) $y = 2 + \frac{3}{1-x}$

$1-x=0 \Rightarrow x=1$



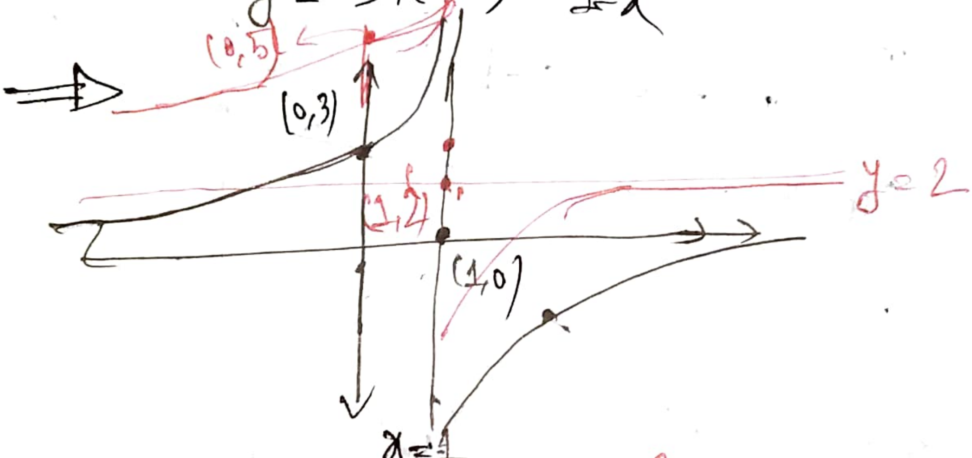
$\Longrightarrow$

$y = f(-x+1) = \frac{1}{1-x}$



$y = 3f(-x+1) = \frac{3}{1-x}$

$\Longrightarrow$

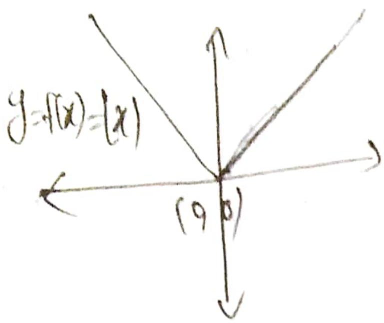


$y = 3f(-x+1) + 2 = 2 + \frac{3}{1-x}$

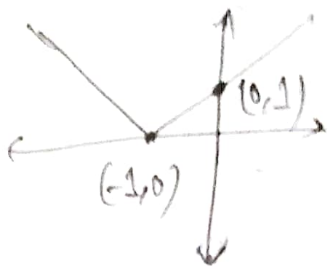
$\text{Dom } f = \mathbb{R} - \{1\}$

$\text{Range } f = \mathbb{R} - \{2\}$

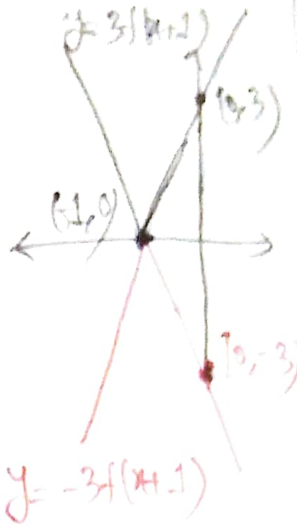
(j) $y = 2 - 3|x+1|$



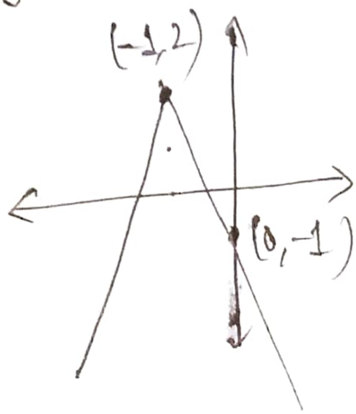
$\Longrightarrow$



$\Longrightarrow$



$\Longrightarrow$ $y = -3f(x+1) + 2$

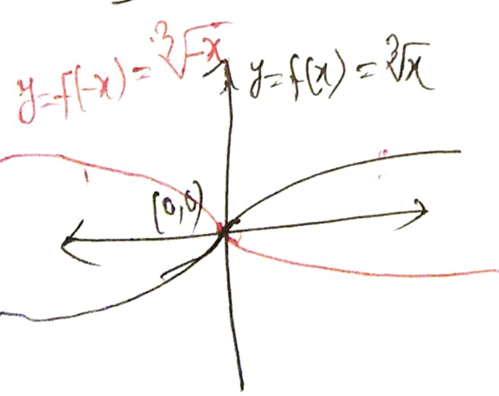


$\boxed{\begin{array}{l} \text{Dom } f = \mathbb{R} \\ \text{Range } f = (-\infty, 2] \end{array}}$

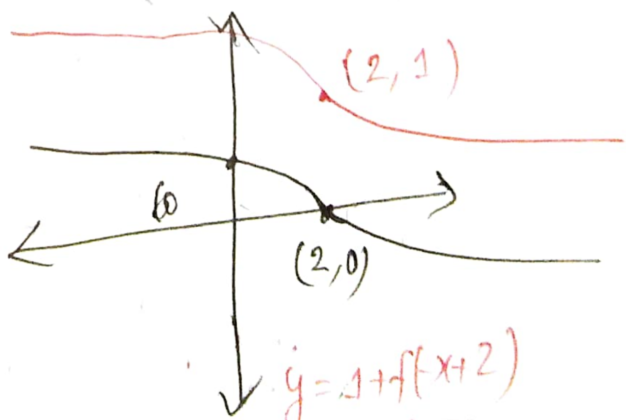
$-x + 2 = 0 \Rightarrow x = 2$

$y = f(-x+2) = \sqrt[3]{2-1}$

(k) $y = 1 + \sqrt[3]{2-x}$



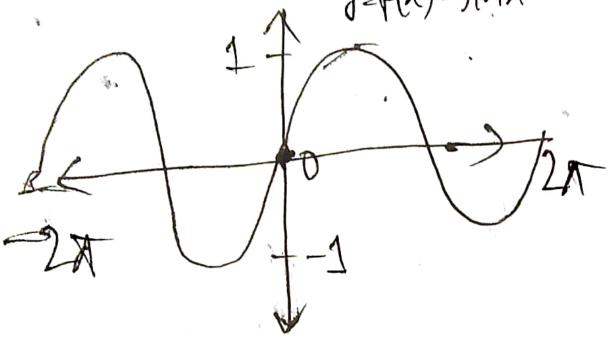
$\Longrightarrow$



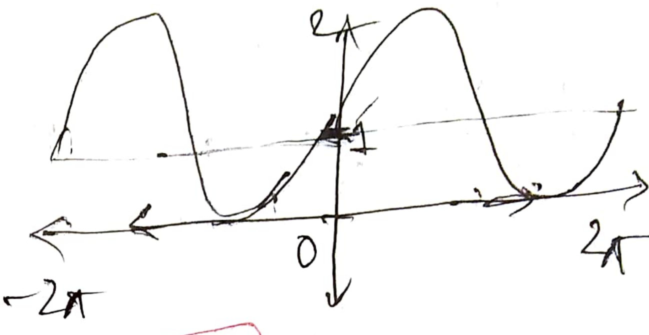
$\boxed{\begin{array}{l} \text{Dom } f = \mathbb{R} \\ \text{Range } f = \mathbb{R} \end{array}}$

(l) $y = 1+\sin x$

$y = f(x) = \sin x$



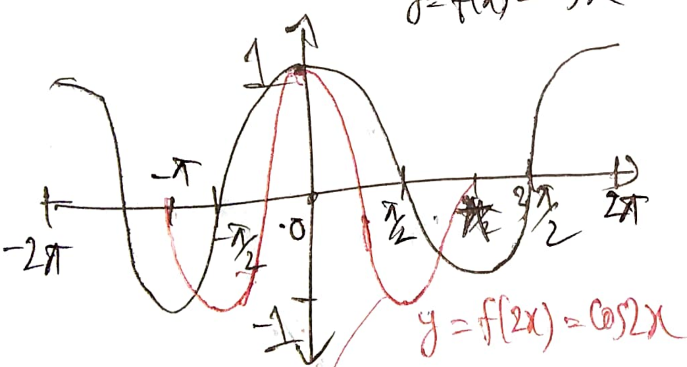
$y = 1+f(x) = 1+\sin x$



$\boxed{\text{Dom} f = \mathbb{R}, \quad \text{Range} f = [0,2]}$

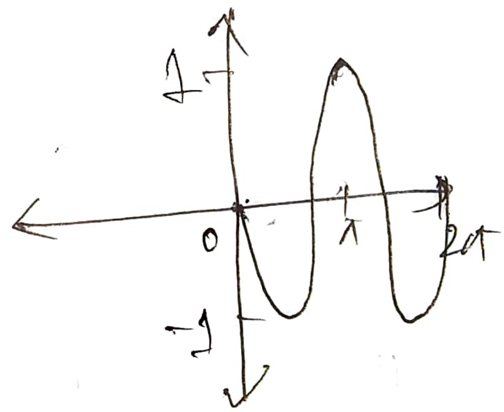
(m) $y = \cos(2x+\pi)$

$y = f(x) = \cos x$



$y = f(2x) = \cos 2x$

$y = f(2x+\pi)$

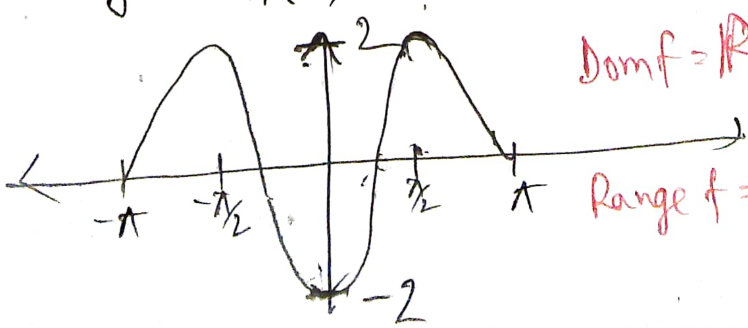


Dom f = $\mathbb{R}$

Range f = $[-1,1]$

$y = -2f(2x) = -2\cos 2x$

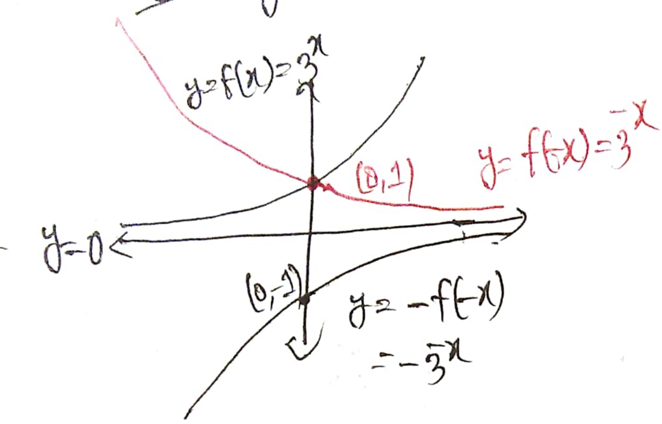
(n)

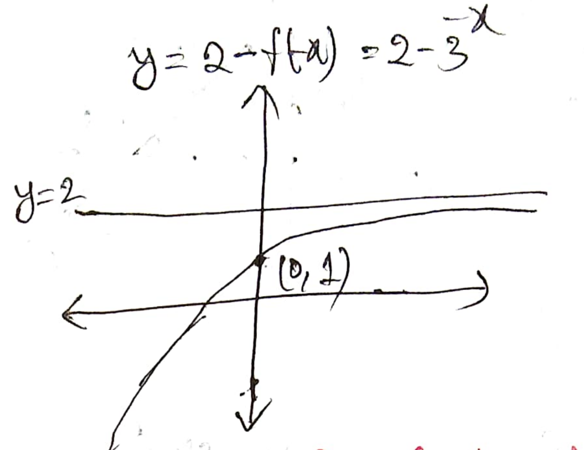


Dom f = $\mathbb{R}$

Range f = $[-2,2]$

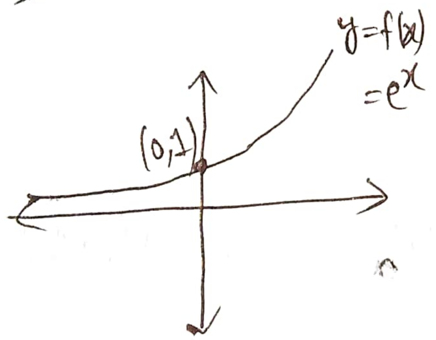
(p) $y = 2 - 3^{-x}$

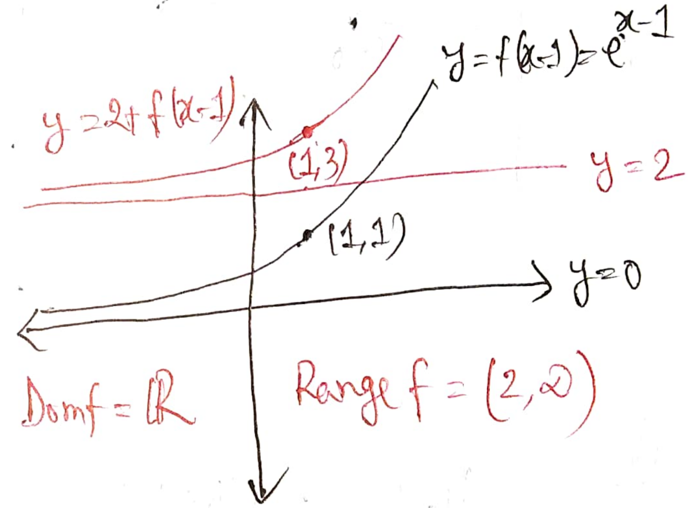




Dom f = $\mathbb{R}$, Range f = $(-\infty, 2)$

(q) $y = 2 + e^{x-1}$

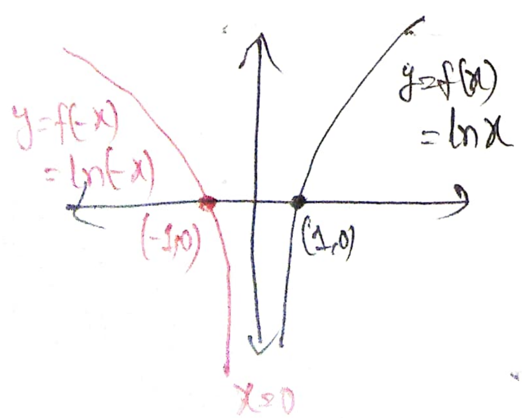


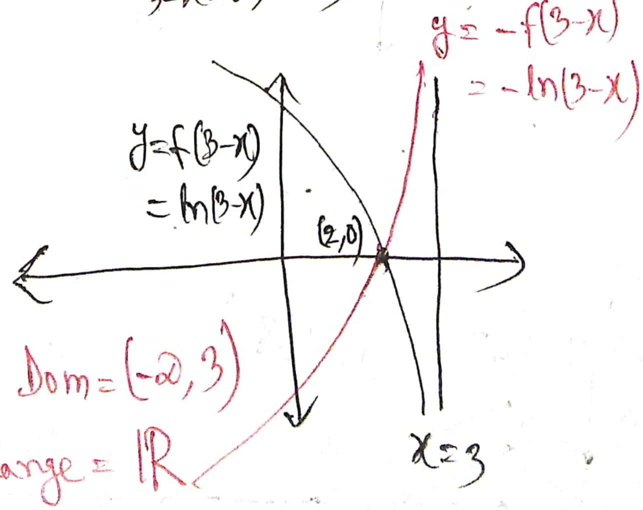


Dom f = $\mathbb{R}$ Range f = $(2, \infty)$

(r) $g(x) = -\ln(3-x)$

$3-x=0 \Rightarrow x=3$

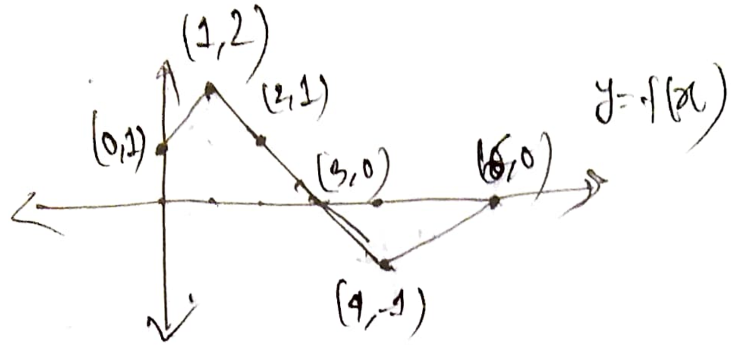




Dom = $(-\infty, 3)$

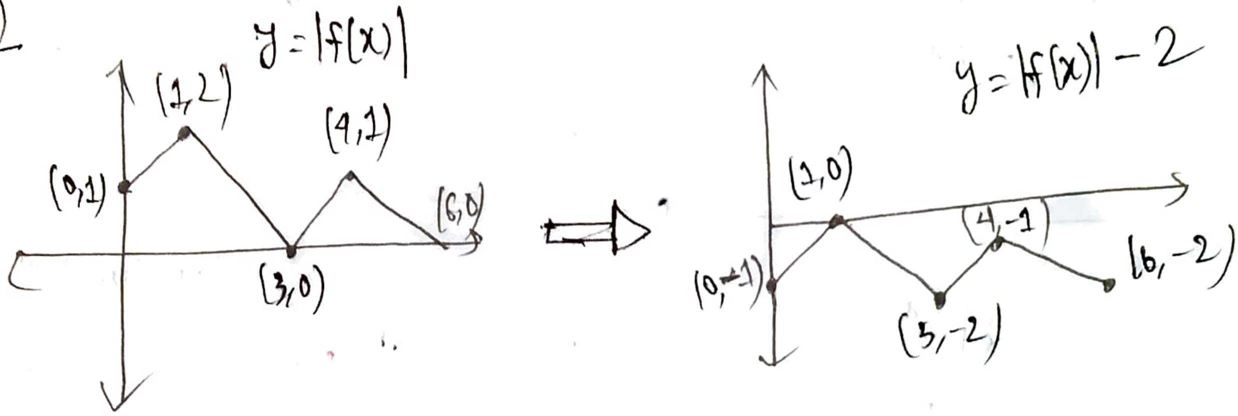
Range = $\mathbb{R}$

7



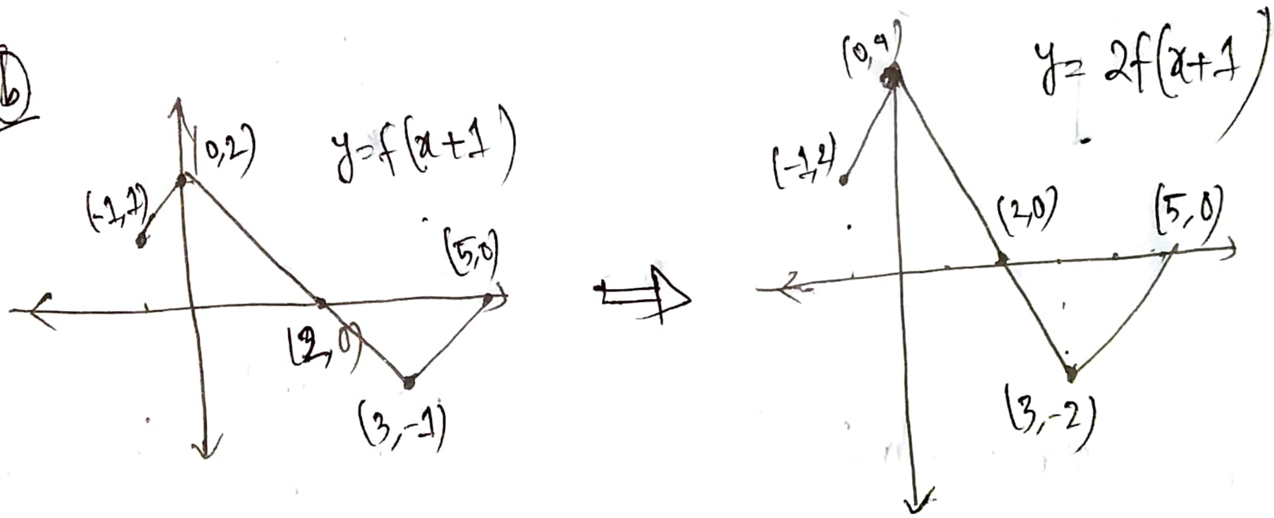
(1,2)
(0,1)
(2,1)
y = f(x)
(3,0)
(6,0)
(4,-1)

(a)



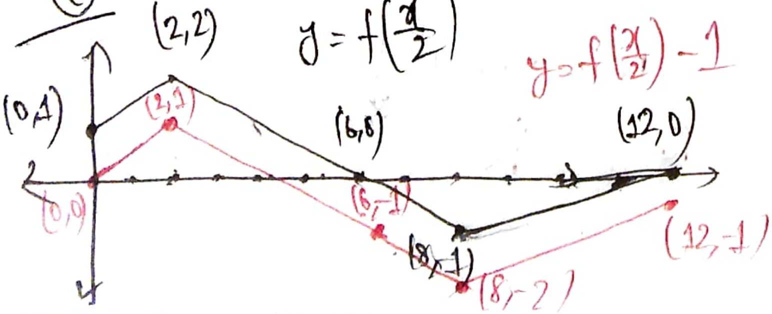
y = |f(x)|
(1,2)
(4,1)
(0,1)
(6,0)
(3,0)
y = |f(x)| - 2
(1,0)
(4,-1)
(0,-1)
(6,-2)
(3,-2)

(b)



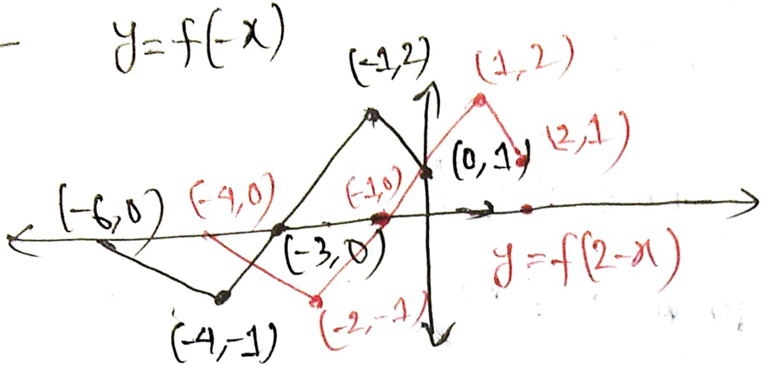
(0,2)
y = f(x+1)
(-1,1)
(5,0)
(2,0)
(3,-1)
(0,4)
y = 2f(x+1)
(-1,2)
(2,0)
(5,0)
(3,-2)

(c)

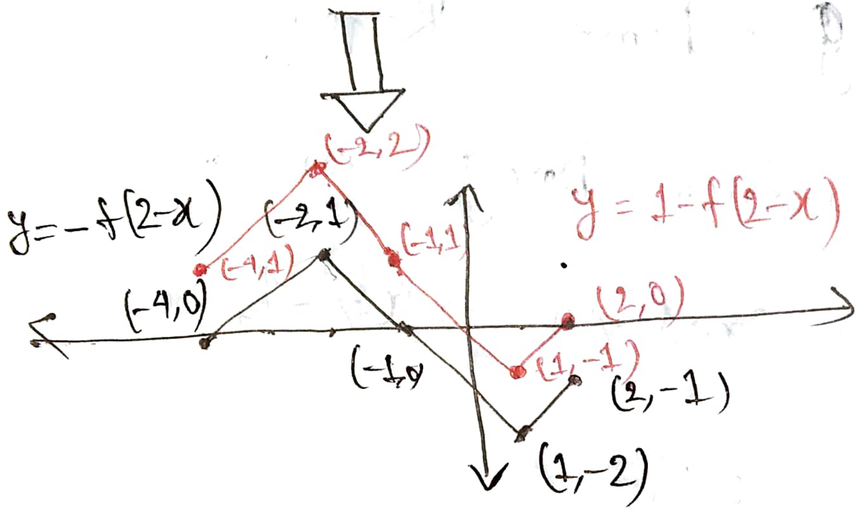


(2,2)
y = f(x/2)
y = f(x/2) - 1
(0,1)
(2,1)
(6,0)
(12,0)
(0,0)
(6,-1)
(8,-1)
(12,-1)
(8,-2)

(d)



$y=f(-x)$
$(-1,2)$
$(1,2)$
$(2,1)$
$(0,1)$
$(-6,0)$
$(-4,0)$
$(-1,0)$
$(-3,0)$
$y=f(2-x)$
$(-4,-1)$
$(-2,-1)$



$(-2,2)$
$y=-f(2-x)$
$(-2,1)$
$(-1,1)$
$y=1-f(2-x)$
$(-4,1)$
$(-4,0)$
$(2,0)$
$(-1,0)$
$(1,-1)$
$(2,-1)$
$(1,-2)$

8(a)

$f(x) = \frac{1}{x}, \quad g(x) = 2x+1$

$\therefore (f \circ g)(x) = f(g(x)) = f(2x+1) = \frac{1}{2x+1}$

Now, for domain $2x+1 \neq 0 \Rightarrow x \neq -\frac{1}{2}$

$\therefore \text{Dom}(f \circ g) = \mathbb{R} - \{-\frac{1}{2}\}$

& $\text{Range}(f \circ g) = \mathbb{R} - \{0\}$ [Note: Similar reasoning to $\frac{1}{x}$]

$\therefore (g \circ f)(x) = g(f(x)) = g\left(\frac{1}{x}\right) = \frac{2}{x} + 1$

$\therefore \text{Dom}(g \circ f) = \mathbb{R} - \{0\}$

Now, $\frac{1}{x} \neq 0 \Rightarrow \frac{2}{x} \neq 0 \Rightarrow \frac{2}{x} + 1 \neq 1$

$\Rightarrow \text{Range}(g \circ f) = \mathbb{R} - \{1\}$

$\therefore (f \circ f)(x) = f(f(x)) = f\left(\frac{1}{x}\right) = \frac{1}{\frac{1}{x}} = x$

$\therefore \text{Dom}(f \circ f) = \mathbb{R}$, $\text{Range}(f \circ f) = \mathbb{R}$

& $(g \circ g)(x) = g(g(x)) = g(2x+1) = 2(2x+1)+1$
$= 4x+2+1$
$= 4x+3$

$\therefore \text{Dom}(g \circ g) = \mathbb{R}$, $\text{Range}(g \circ g) = \mathbb{R}$

(b) $f(x) = \sqrt{x}$, $g(x) = \sqrt{2-x}$

$\therefore (f \circ g)(x) = f(g(x)) = f(\sqrt{2-x}) = \sqrt{\sqrt{2-x}} = (2-x)^{1/4}$

For, domain, $2-x \geq 0 \Rightarrow 2 \geq x$

$\therefore \text{Dom}(f \circ g) = (-\infty, 2]$
$\text{Range}(f \circ g) = [0, \infty)$

$\therefore (g \circ f)(x) = g(f(x)) = g(\sqrt{x}) = \sqrt{2-\sqrt{x}}$

For Domain,

$x \geq 0$ and $2 - \sqrt{x} \geq 0$

$\hookrightarrow$ (i) $\quad \Rightarrow 2 \geq \sqrt{x}$

$\therefore 4 \geq x \longrightarrow$ (ii)

Combining (i) & (ii), [Note; taking the common part]

$\text{Dom}(g \circ f) = (\infty, 4]$

& $\text{Range}(g \circ f) = [0, \infty)$

$\therefore (f \circ f)(x) = f(f(x)) = f(\sqrt{x}) = \sqrt{\sqrt{x}} = x^{1/4}$

$\hookrightarrow x \geq 0$

$\therefore \text{Dom}(f \circ f) = [0, \infty)$

$\text{Range}(f \circ f) = [0, \infty)$

$\therefore (g \circ g)(x) = g(g(x)) = g(\sqrt{2-x})$

$= \sqrt{2-\sqrt{2-x}}$

For domain,

$2-x \geq 0$ and $2-\sqrt{2-x} \geq 0$

$\Rightarrow 2 \geq x \longrightarrow$ (i)

$\Rightarrow 2 \geq \sqrt{2-x}$

$\Rightarrow 4 \geq 2-x$

$\Rightarrow \cancel{x} \geq 2-4$

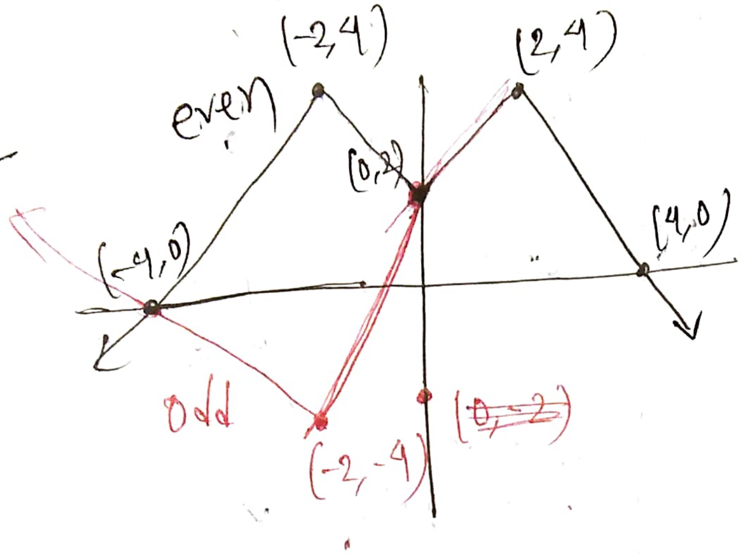
$\therefore x \geq -2 \longrightarrow$ (ii)

Combining (i) & (ii) $\Rightarrow -2 \leq x \leq 2$

$\therefore$ Dom(gog) $= [-2, 2]$

Range(gog) $= [0, 2)$

9)



10) (a) $y = f(x) = \sqrt[3]{x} - 1 \Rightarrow f(-x) = \sqrt[3]{-x} - 1 = -\sqrt[3]{x} - 1$

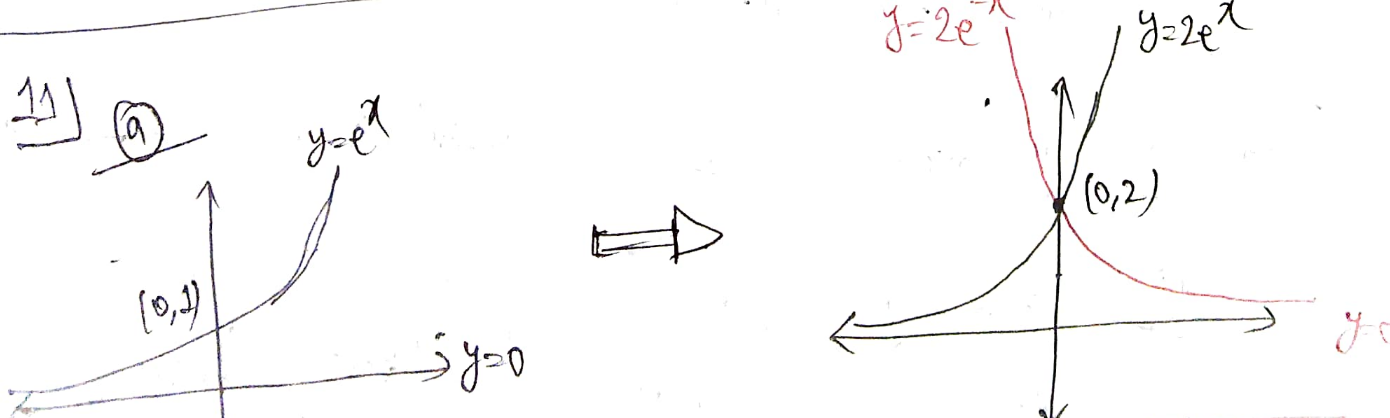
$\boxed{\text{Neither}}$

(b) $y = f(x) = \dfrac{x^3}{x^4 - 1}$; $f(-x) = \dfrac{(-x)^3}{(-x)^4 - 1} = \dfrac{-x^3}{x^4 - 1} = -f(x)$

$\boxed{\text{Odd}}$

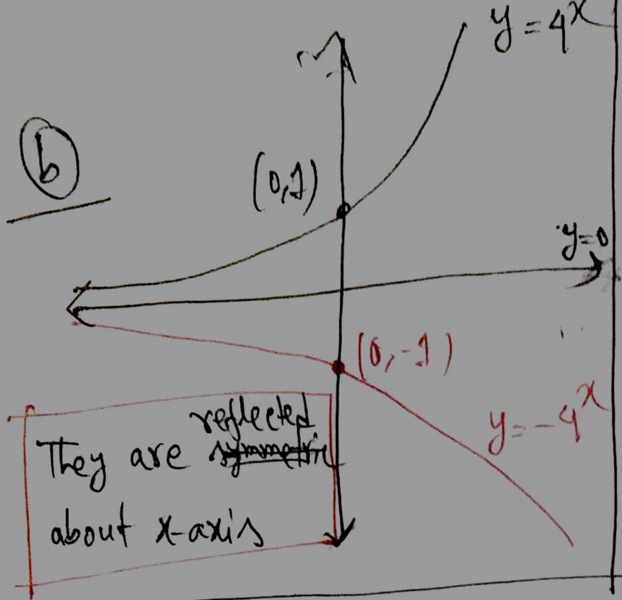
(c) $y = f(x) = \dfrac{2}{x-3}$; $f(-x) = \dfrac{2}{-x-3}$ $\boxed{\text{Neither}}$

11) (a)

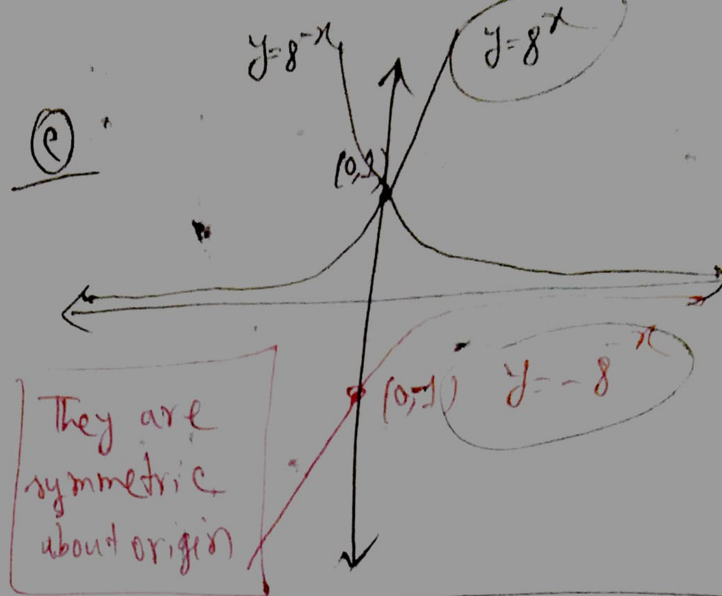


They are ~~symmetric~~ reflected about y-axis

(b)



(c)



12)

x	-3	-2	-1	0	1	2	3
(a) (even) $f(x)$	1	-5	-1	0	-1	-5	1
(b) (odd)		5			1		-1

13) (a) $y = 5x + C$ (b) $y - 5 = m(x+2)$

(c)
$$\left.\begin{aligned} m=0 &\Rightarrow y=5 \\ m=1 &\Rightarrow y=x+7 \\ m=-1 &\Rightarrow y=-x+3 \end{aligned}\right\} \text{ sketch them yourselves}$$

14) (a) $y = mx + 3$ (b) $y = \tan 150^\circ \cdot x + C = -\frac{1}{\sqrt{3}}x + C$

(c)
$$\left.\begin{aligned} C=0, &\quad y = -\tfrac{1}{\sqrt{3}}x \\ C=1, &\quad y = -\tfrac{1}{\sqrt{3}}x + 1 \\ C=-2, &\quad y = -\tfrac{1}{\sqrt{3}}x - 2 \end{aligned}\right\} \text{ Sketch them yourselves}$$

15| $y = 5x$ (Red), $y = \frac{x}{5}$ (Green Straight line)

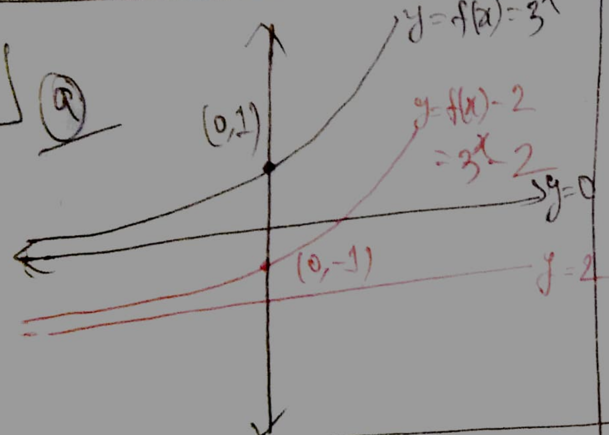
$y = x^5$ (Purple), $y = \sqrt[5]{x}$ (Black), $y = 1+e^x$ (Green Curve)

and $y = \ln(x-1)$ (Blue)

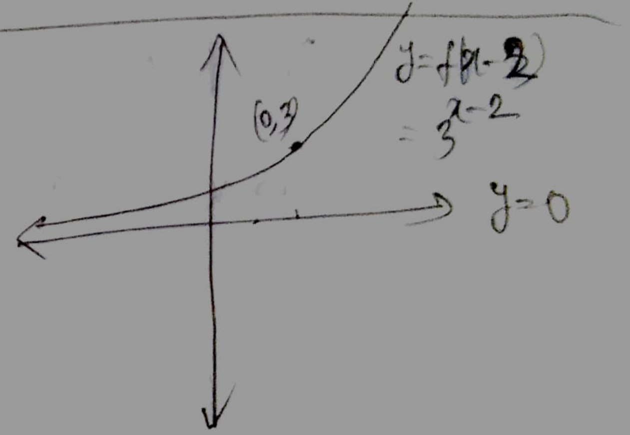
Inverse: $y = 5x$ & $y = \frac{x}{5}$

$y = x^5$ & $y = \sqrt[5]{x}$; $y = 1+e^x$ & $y = \ln(x-1)$

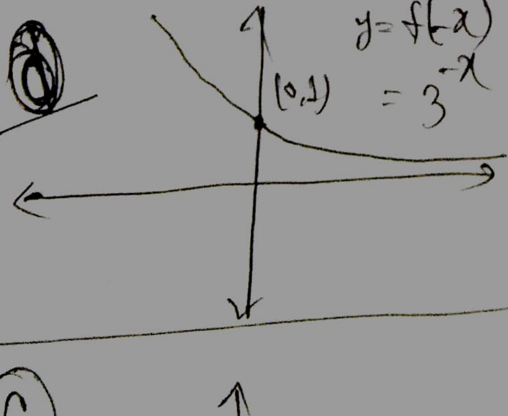
16| (a)



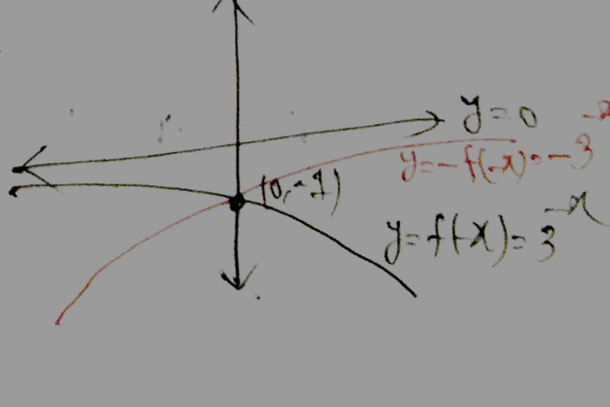
(b)



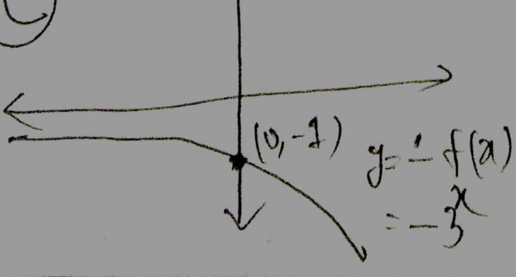
(d)



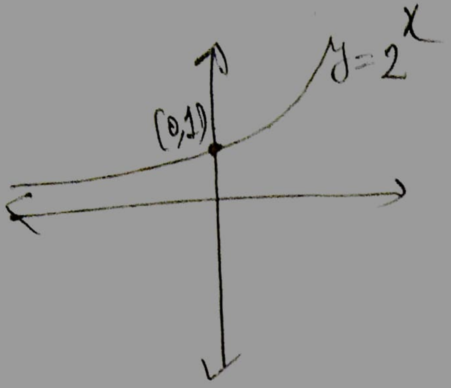
(e)



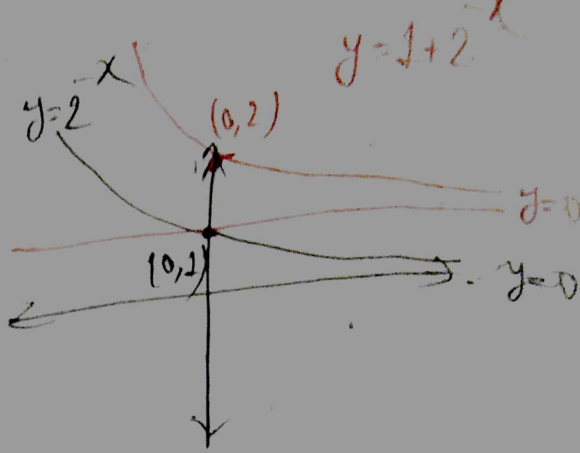
(c)



17| (a) $f(x) = 1 + 2^{-x}$



$\Longrightarrow$



By horizontal line test (HLT),

The function is one - one.

$\therefore$ Inverse function exists.

Here $y = f(x) = 1 + 2^{-x}$

$\Rightarrow y - 1 = 2^{-x}$

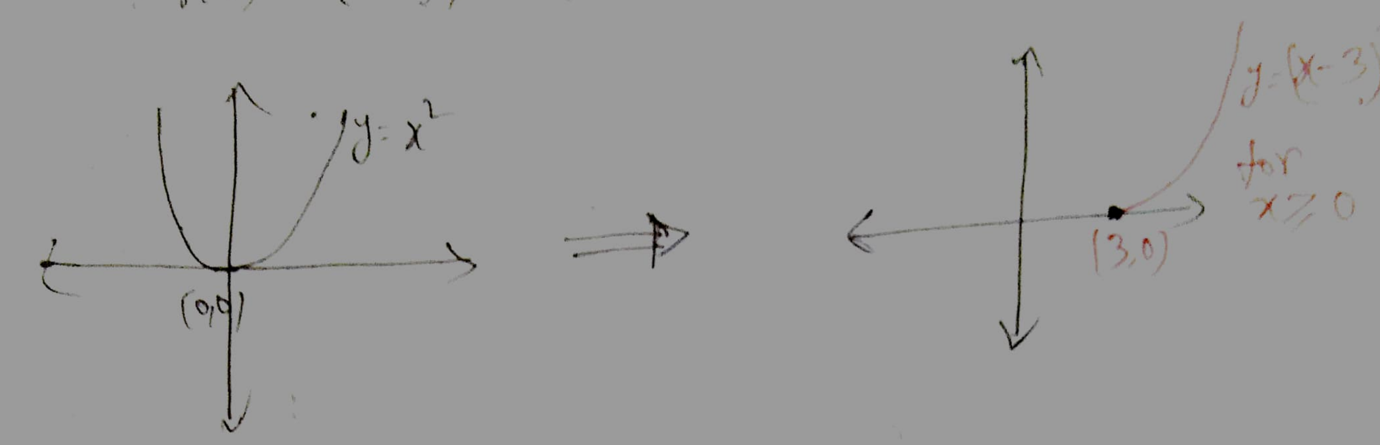
$\Rightarrow \log_2 (y-1) = -x$

$\Rightarrow x = -\log_2 (y-1)$

$\Rightarrow f^{-1}(y) = -\log_2 (y-1)$

$\therefore \boxed{f^{-1}(x) = -\log_2 (x-1)}$

(b) $f(x) = (x-3)^2;\ x \geq 3$



Thus, by HLT $f(x)$ is one to one.

Now, $y = f(x) = (x-3)^2$

$\Rightarrow \sqrt{y} = x - 3 \ ;[\because x \geq 3]$

$\Rightarrow x = \sqrt{y} + 3 \Rightarrow f^{-1}(y) = \sqrt{y} + 3 \ \therefore \boxed{f^{-1}(x) = \sqrt{x} + 3}$

---

18) (a) $y = f(x) = (x-2)^3 + 1$

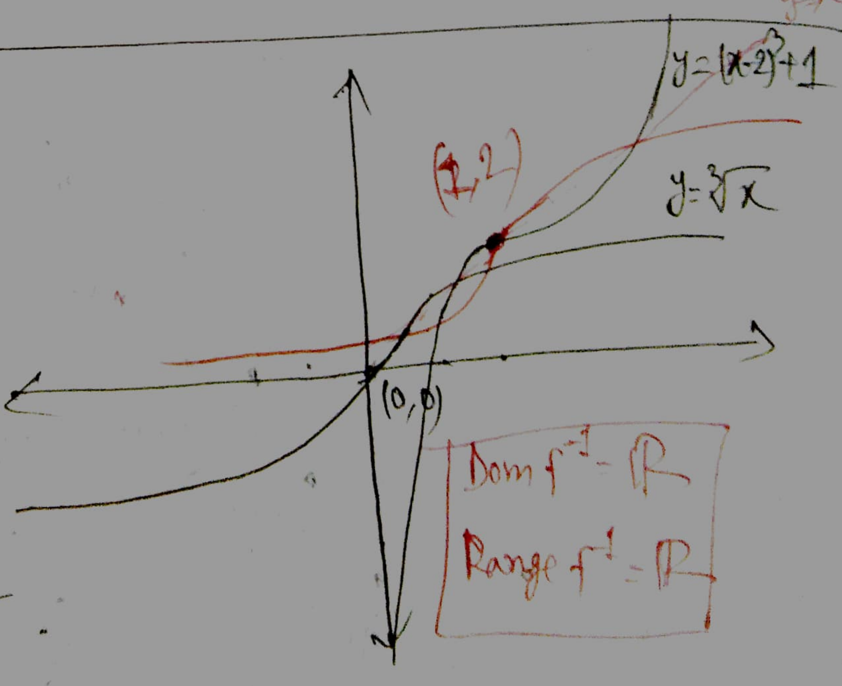
$\Rightarrow (x-2)^3 = y - 1$

$\Rightarrow x - 2 = \sqrt[3]{y-1}$

$\Rightarrow x = \sqrt[3]{y-1} + 2$

$\Rightarrow f^{-1}(y) = \sqrt[3]{y-1} + 2$

$\therefore f^{-1}(x) = \sqrt[3]{x-1} + 2$



$\boxed{\begin{array}{l} \text{Dom } f^{-1} = \mathbb{R} \\ \text{Range } f^{-1} = \mathbb{R} \end{array}}$

(b) $y = f(x) = 2 + e^{-x}$

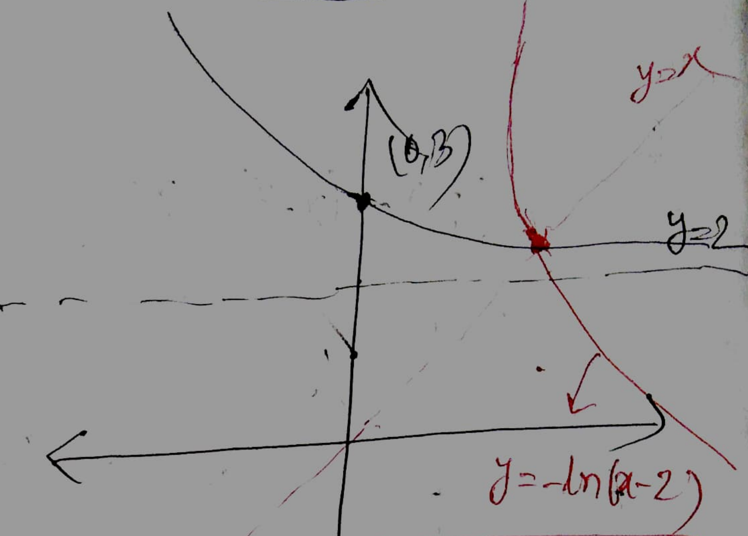
$\Rightarrow e^{-x} = y - 2$

$\Rightarrow -x = \ln(y-2)$

$\Rightarrow x = -\ln(y-2)$

$\Rightarrow f^{-1}(y) = -\ln(y-2)$

$\therefore \boxed{f^{-1}(x) = -\ln(x-2)}$



$x > 2$

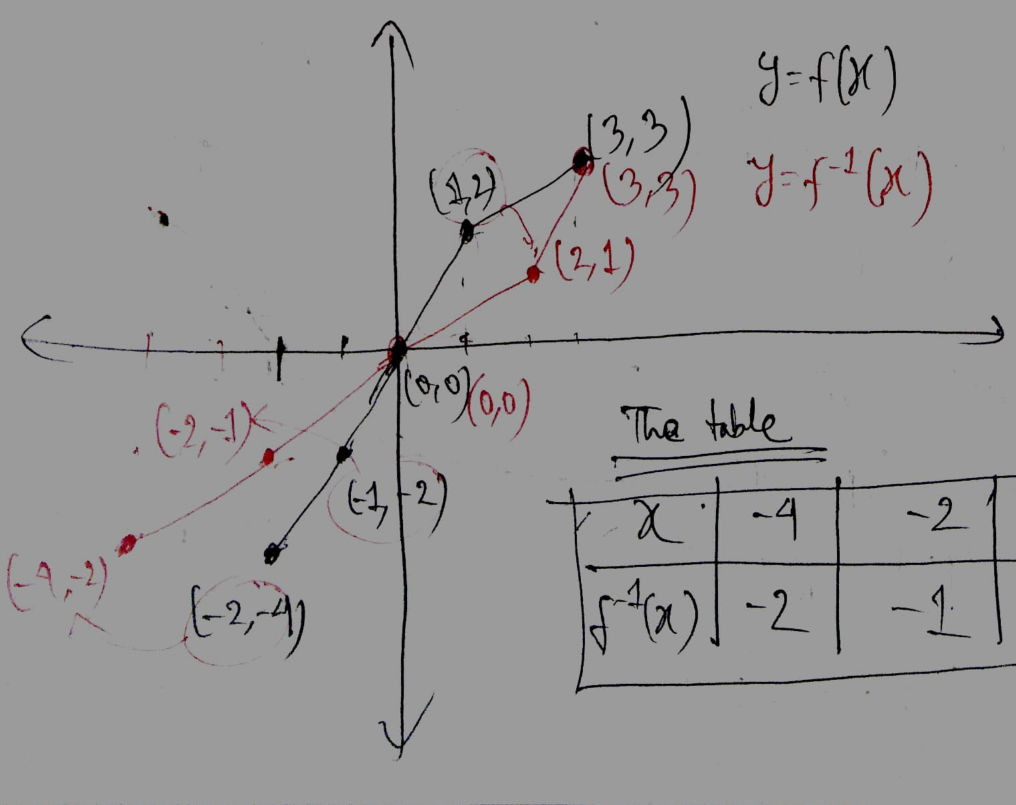
$\text{Dom} f^{-1} = (2, \infty)$

$\text{Range} f^{-1} = \mathbb{R}$

---

19) Given,

x	-2	-2	2	3
$f(x)$	-4	-2	2	3



The table

x	-4	-2	2	3
$f^{-1}(x)$	-2	-1	1	3

20) (a) $f(x) = \sqrt[5]{x+2}$, $g(x) = x^5 - 2$

Here, $y = f(x) = \sqrt[5]{x+2}$

$\Rightarrow y^5 = x+2$

$\Rightarrow x = y^5 - 2 \Rightarrow f^{-1}(y) = y^5 - 2) \Rightarrow f^{-1}(x) = x^5 - 2 = g(x)$

i.e. they are inverse

(b) $f(x) = x^4$, $g(x) = \sqrt[4]{x}$

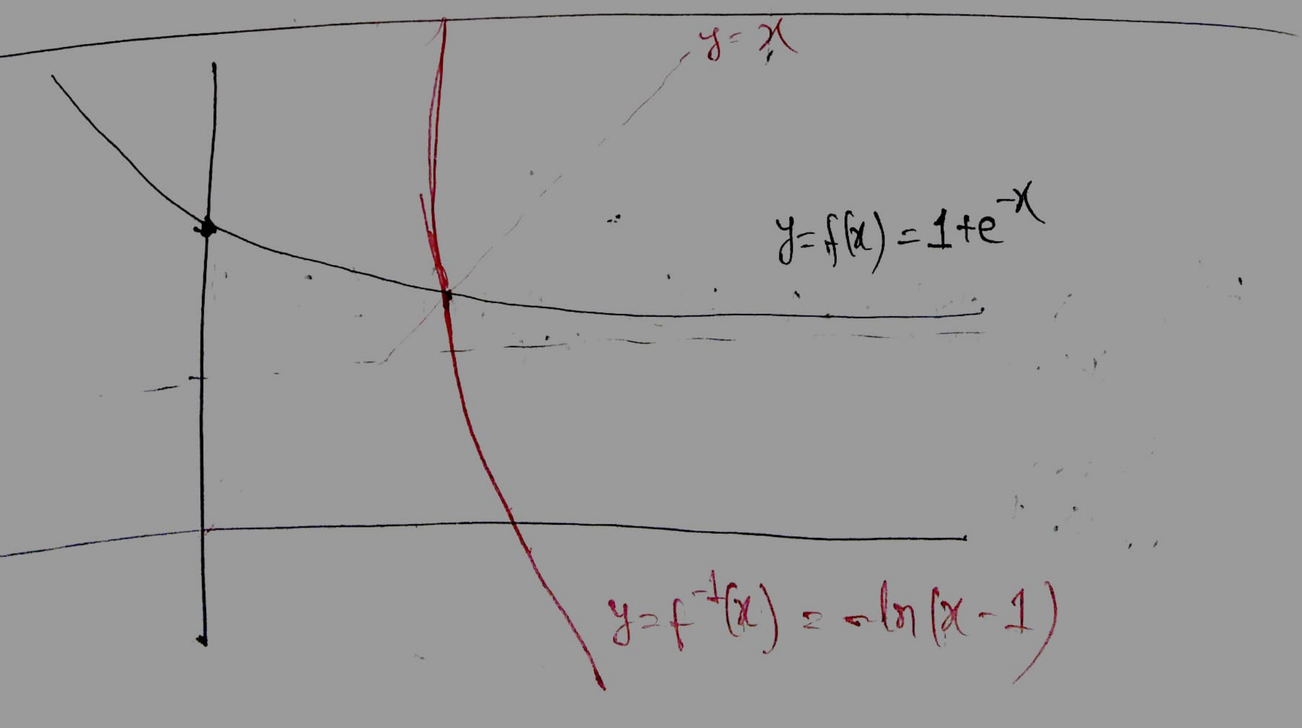
Here, $y = f(x) = x^4$

$\Rightarrow x = \pm \sqrt[4]{y}$

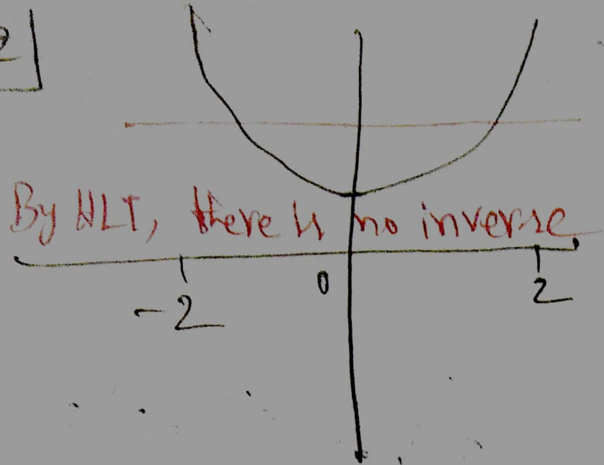
$\Rightarrow f^{-1}(y) = \pm \sqrt[4]{y} \Rightarrow f^{-1}(x) = \pm \sqrt[4]{x} \neq g(x)$

i.e. they are not inverse

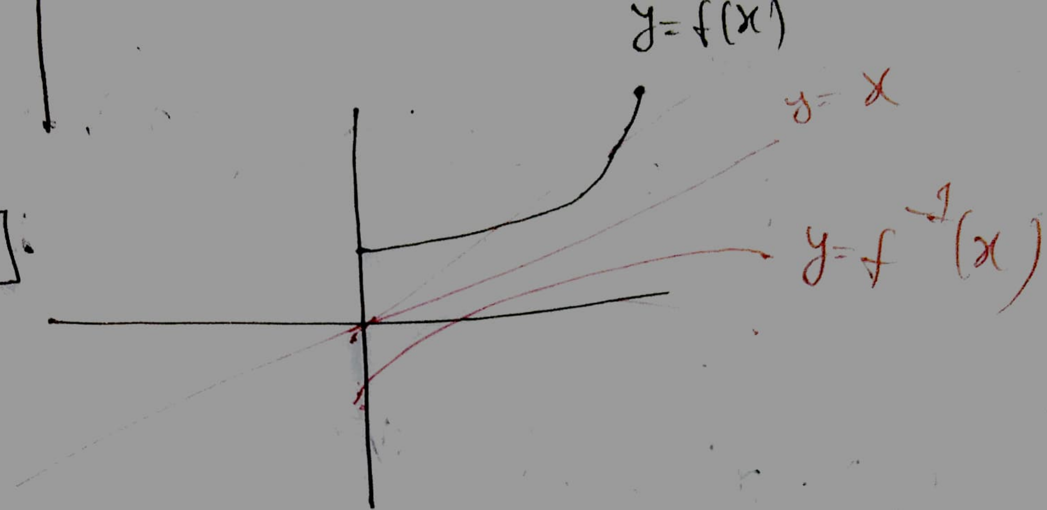
21)



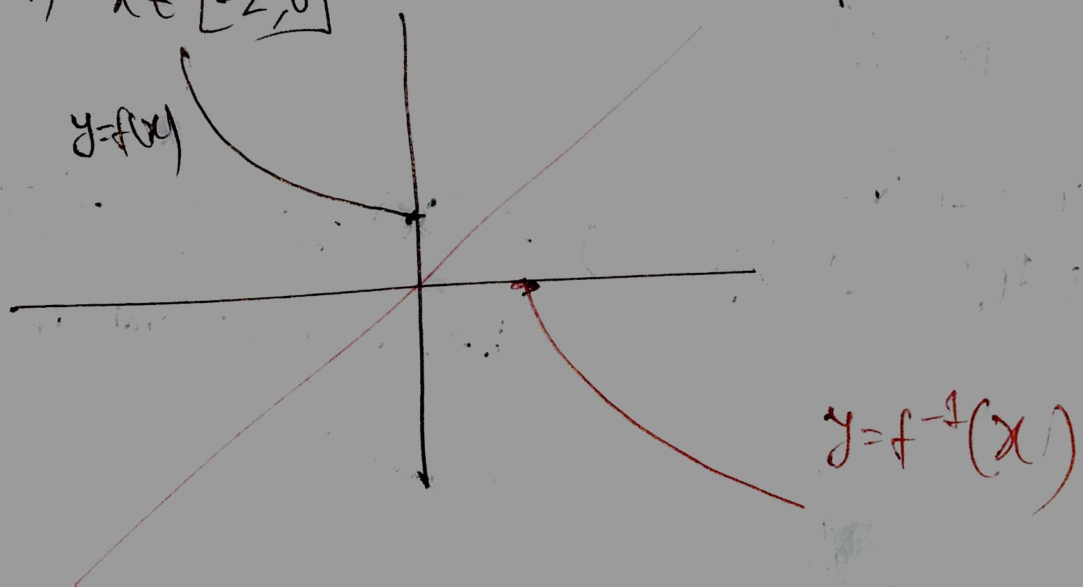
22|



If $x \in [0,2]$:



or, $x \in [-2,0]$



23| (a) $\lim_{x \to -1^-} f(x) = -1$; (b) $\lim_{x \to 0^+} f(x) = 1$; (c)

(c)

$LHL = \lim_{x \to -1^-} f(x) = -1$

$RHL = \lim_{x \to -1^+} f(x) = 0$ $\therefore \lim_{x \to -1} f(x) = DNE$

(d) $LHL = RHL$; $\lim_{x \to 1} = 0$

(e)

$f(-2) =$ not defined

$f(-1) = -1$, $f(2) = -1$, $f(3) = 0$

(f)

$LHL = \lim_{x \to -2^-} f(x) = -\infty$

$RHL = \lim_{x \to -2^+} f(x) = +\infty$

$f(-2) =$ not defined

$\therefore$ Discontinuous at $x = -2$

$LHL = \lim_{x \to 0^-} f(x) = 1$

$RHL = \lim_{x \to 0^+} f(x) = 1$

$f(0) = 1$

$\therefore$ Continuous at $x = 0$..

$LHL = \lim_{x \to 2^-} f(x) = -2$

$RHL = \lim_{x \to 2^+} f(x) = -2$

$f(2) = -1$

$\therefore$ Discontinuous at $x = 2$

(g) VA

$\lim_{x \to -2^-} f(x) = -\infty$; $\lim_{x \to -2^+} f(x) = \infty$

& $f(-2) =$ not defined

$\therefore x = -2$ is a VA

HA

$\lim_{x \to -\infty} f(x) = 3$ But, there is some value of x such that $f(x) = 3$

Also, $\lim_{x \to \infty} f(x) = 1$

But $f(0) = 1$ Thus, no HA

24)

$f(x) = 2 - |x-1|$

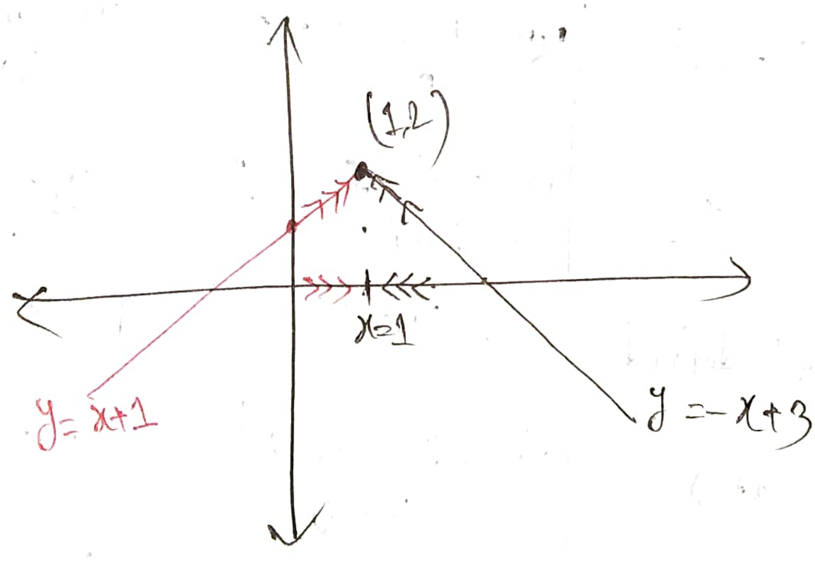
$$= \begin{cases} 2-(x-1) \text{ or, } -x+3 & ; \; x-1 \geq 0 \text{ or, } x \geq 1 \\ 2-\{-(x-1)\} \text{ or, } x+1 & ; \; x-1 < 0 \text{ or } x < 1 \end{cases}$$

Let, $f_1(x) = -x+3$; $f_2(x) = x+1$

$f_1(1) = -1+3 = 2$; $f_2(1) = 1+1 = 2$

$\therefore$ $f(x)$ is cont^n at $x = 1$

or, sketch the graph



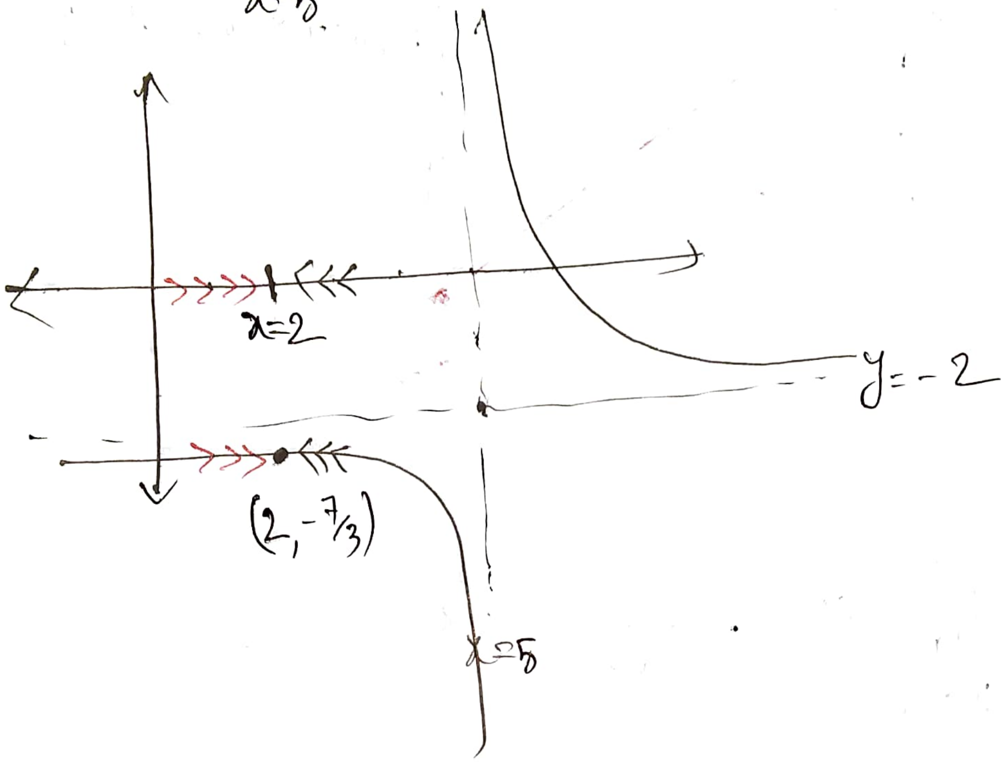
$\text{LHL} = \lim_{x \to 1^-} f(x) = 2$

$\text{RHL} = \lim_{x \to 1^+} f(x) = 2$ & $f(1) = 2$

Thus, cont^n at $x = 1$

25

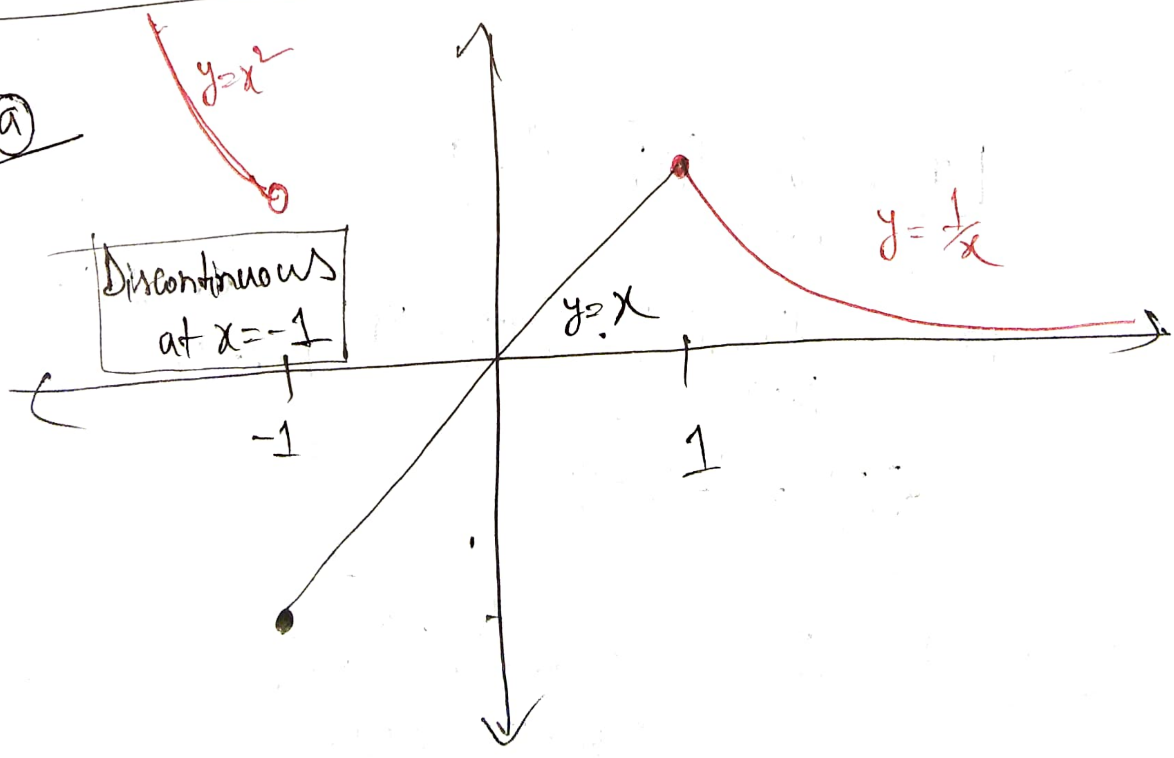
$$f(x) = \frac{1}{x-5} - 2$$



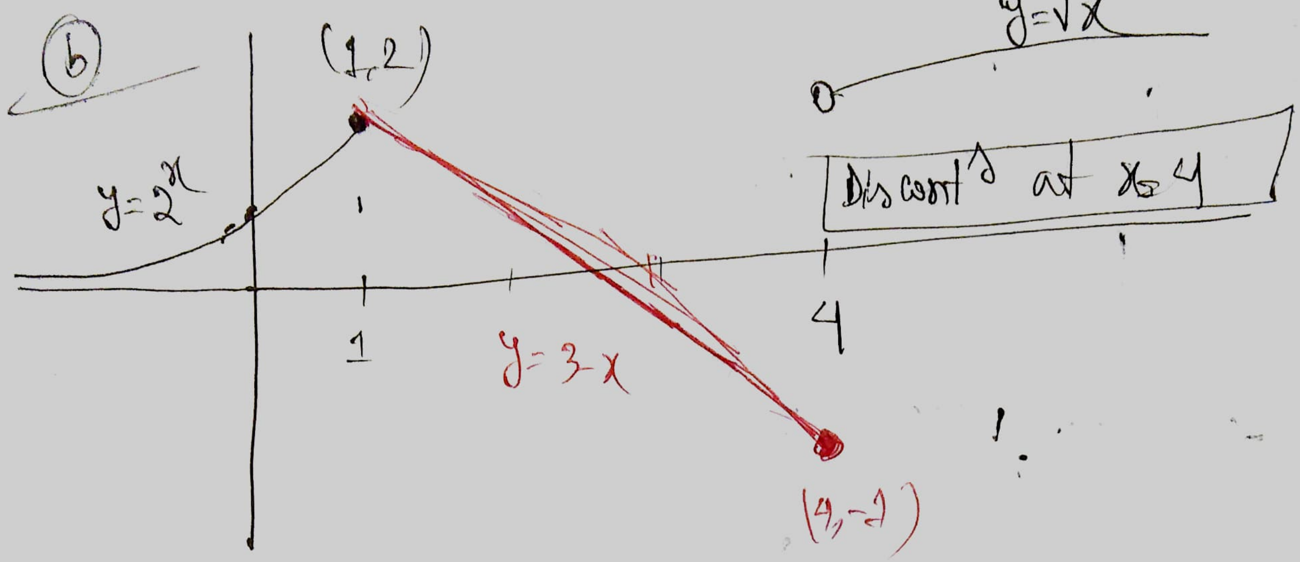
LHL $= \lim_{x \to 2^-} f(x) = -\frac{7}{3}$; RHL $= \lim_{x \to 2^+} f(x) = -\frac{7}{3}$; $f(2) = \frac{1}{2-5} - 2 = -\frac{7}{3}$

Thus, f(x) is cont^n at x = 2

26 (a)



(b)



27] (a) $LHL = \lim_{x \to -2^-} f(x) = -2$

$RHL = \lim_{x \to -2^+} f(x) = 2^2 - 5 = -1$

$\therefore \lim_{x \to -2} f(x) = DNE$

(b)

$LHL = \lim_{x \to 3^-} f(x) = 3^2 - 5 = 4$

$RHL = \lim_{x \to 3^+} f(x) = \sqrt{3+13} = 4$

~~$f(3) = 3^2 - 5 = 4$~~

$\lim_{x \to 3} f(x) = 4$

28] & 29] Check Recorded Class

01NOV & 02NOV